

|                                 |                                                                                             |
|---------------------------------|---------------------------------------------------------------------------------------------|
| <b>EX NO:</b><br><b>8a</b>      | <b>SIMULATION AND ANALYSIS OF BPSK MODULATION USING<br/>CONSTITUTION DIAGRAMS IN MATLAB</b> |
| <b>DATE:1</b><br><b>0/08/26</b> |                                                                                             |

### Objective

- i) To generate BPSK symbols from a binary data sequence using phase-shift keying principles.
- ii) To Plot and analyze the BPSK signal constellation in the In-Phase (I) and Quadrature (Q) plane.
- iii) To investigate the effect of noise on constellation points and evaluate the resulting degradation in symbol detection performance.

### Matlab code

```

clc;
clear;
close all;
%% Objective 1: Generate BPSK symbols
N = 1000;                % Number of bits
data = randi([0 1], N, 1); % Random binary sequence
% BPSK Mapping
bpsk_symbols = 2*data - 1;
%% Objective 2: Plot BPSK Signal Constellation
figure;
subplot(1,2,1);
scatter(real(bpsk_symbols), imag(bpsk_symbols), ...
        30, 'filled');
hold on;
xline(0,'k','LineWidth',1.5);
yline(0,'k','LineWidth',1.5);
grid on;
axis equal;
xlim([-1.5 1.5]);
ylim([-1 1]);
xlabel('In-Phase (I)');
ylabel('Quadrature (Q)');
title('Ideal BPSK Constellation');
%% Objective 3: Add Noise and Evaluate Performance
SNR = 5;                % Signal-to-Noise Ratio (dB)
rx_signal = awgn(bpsk_symbols, SNR, 'measured');

subplot(1,2,2);

```

```

scatter(real(rx_signal), imag(rx_signal), ...
        15, 'filled');
hold on;
xline(0,'k','LineWidth',1.5);
yline(0,'k','LineWidth',1.5);
grid on;
axis equal;
xlim([-2.5 2.5]);
ylim([-2 2]);
xlabel('In-Phase (I)');
ylabel('Quadrature (Q)');
title(['Noisy BPSK Constellation (SNR = ', num2str(SNR), ' dB)']);
%% Symbol Detection
detected_bits = real(rx_signal) > 0;
% Calculate Errors
num_errors = sum(data ~= detected_bits);
BER = num_errors/N;
fprintf('Number of Transmitted Bits : %d\n', N);
fprintf('Number of Bit Errors          : %d\n', num_errors);
fprintf('Bit Error Rate (BER)          : %f\n', BER);

```

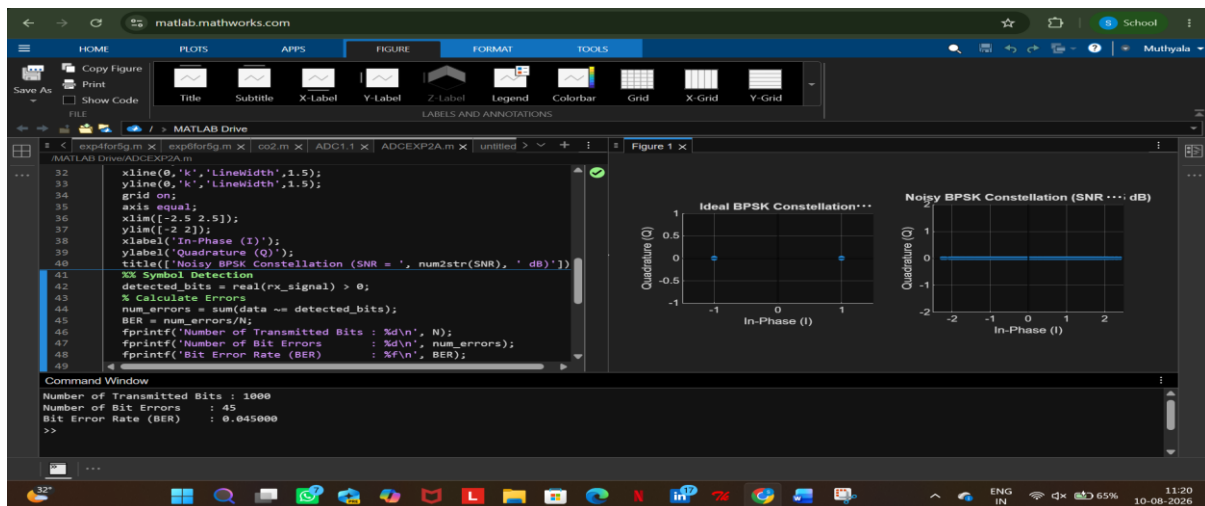
## Procedure:

### Objective 1: Generate BPSK Symbols from a Binary Data Sequence Using Phase-Shift

#### Keying Principles

1. Initialize the MATLAB environment and define the number of bits.
2. Generate a random binary sequence consisting of 0s and 1s.
3. Apply BPSK modulation by mapping binary 0 to  $-1$  and binary 1 to  $+1$ .
4. Store the generated BPSK symbols for further processing.
5. Verify that the symbol sequence contains only two distinct symbol values.
6. Prepare the modulated symbols for constellation analysis.

## OUTPUTGraph



**Result** BPSK symbols were successfully generated from the binary data sequence using phase-shift keying modulation.